

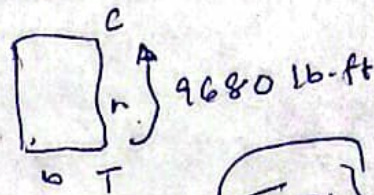
Design the beam

$$M = 9680 \text{ lb-ft}$$

$$\sigma = \frac{Mc}{I}$$

$$I = \frac{1}{12} (bh^3)$$

$$c = \frac{h}{2}$$



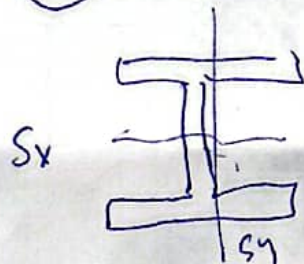
$$\sigma_{all} = 12 \text{ ksi}$$

$$\sigma_{yield} = 36 \text{ ksi}$$

$$F.S. = 3$$

$$h = 2w$$

WF wide flange beam.



$$\sigma = \frac{M}{S}$$

$$\text{section modulus} = S$$

$$S = \frac{I}{c}$$

mm  
in

$$12 \text{ ksi} = \frac{9680 \text{ lb-ft} \cdot 12 \text{ ft}}{S}$$

$$S = 9.68 \text{ in}^3$$

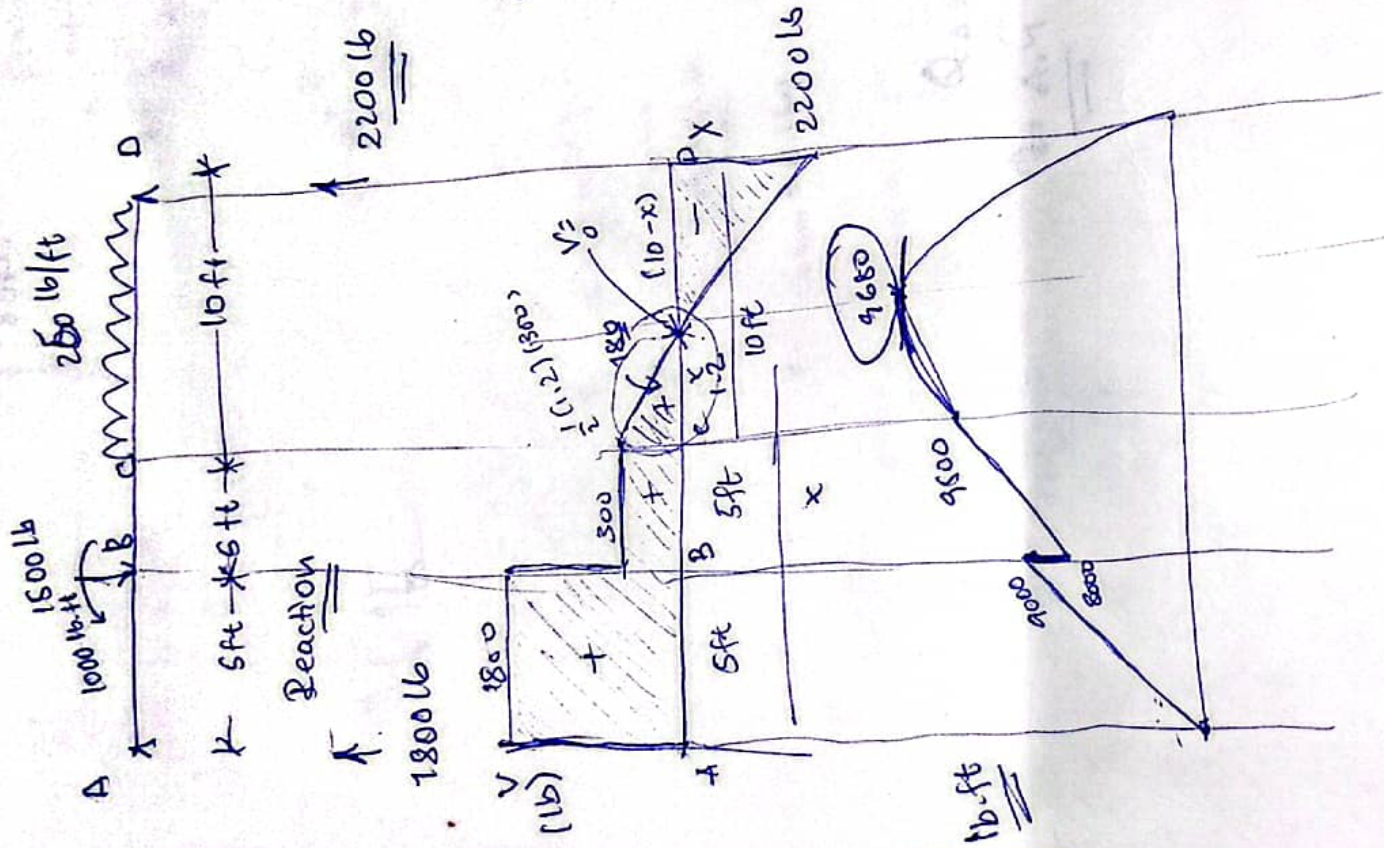
W5 x 19

5 inch = height  
19 lb/ft

W6 x 16

W8 x 13

W10 x



$$V_x = 300 - 250(x-10)$$

$$M_x = -1800x + 160(x-5) + 1000 + 250(x-10)(x-10)$$



$$\frac{300}{250} = \frac{x}{10-x}$$

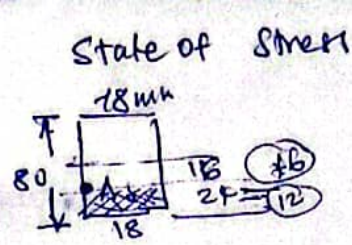
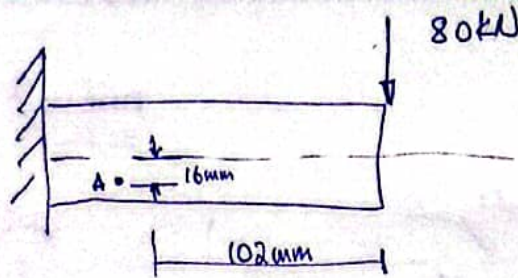
$$\frac{300}{250} = \frac{x}{10-x}$$

$$2200x = 3000 - 300x$$

$$2500x = 3000$$

$$12.5 \times 250 = 3125$$

$$x = 12.5$$



$$\sigma = \frac{P}{A}$$

$$\sigma_x = - \frac{M y}{I}$$

$$\tau_{xy} = - \frac{V Q}{I t}$$

$$I = 0.768 \times 10^{-6} \text{ m}^4$$

$$M = 80 \times 102 \text{ mm} = 8160 \text{ N}\cdot\text{m}$$

$$y = 16 \text{ mm} = 0.016 \text{ m}$$

$$\sigma_x = -170 \text{ MPa}$$

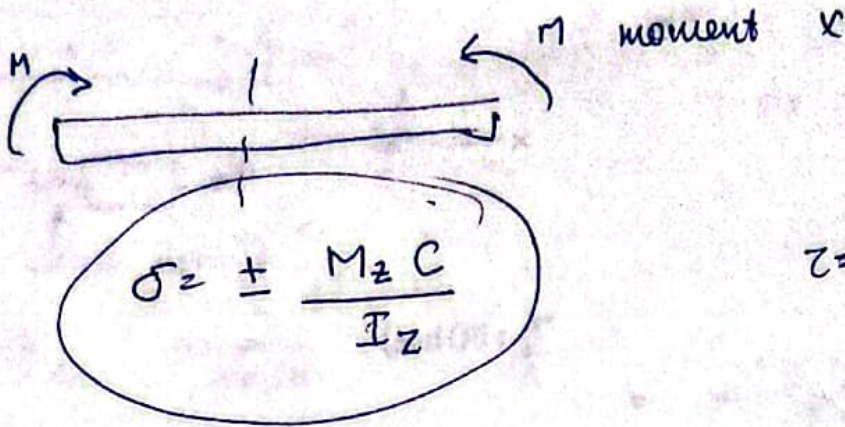
$$I = \frac{b h^3}{12}$$

$$\tau_{xy} = -70 \text{ MPa}$$

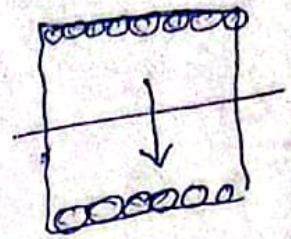
$$Q_A = 18 \times 24$$

$$Q = A \cdot \bar{y} = (18 \times 24)(28) =$$

# Bending

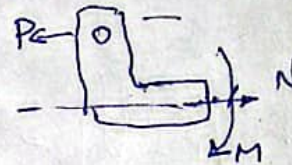
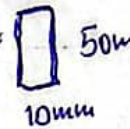
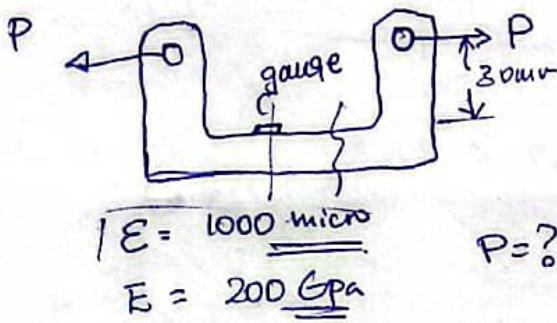


$$\tau = \frac{VQ}{It}$$



$$\sigma_x = + \frac{P}{A} + \frac{M_z C}{I_z} + \frac{M_y C}{I_y}$$

$$\tau_{xy} = - \frac{TC}{J} - \frac{VQ}{It}$$



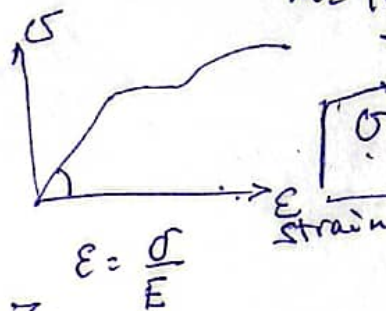
$$\frac{25+30}{2} = \frac{55}{2} = 27.5$$

$$M_z = (0.03)(0.025)P$$

$$\sigma = \epsilon E$$

$$= 200 \times 10^9 \times 1000 \times 10^{-6}$$

$$\sigma = 200 \text{ MPa}$$



$$\sigma_z = + \frac{P}{A} + \frac{M_z C}{I_z}$$

$$A = (10 \times 50) = 500 \text{ mm}^2$$

$$= 500 \times 10^{-6} \text{ m}^2$$

$$I = \frac{bh^3}{12} =$$

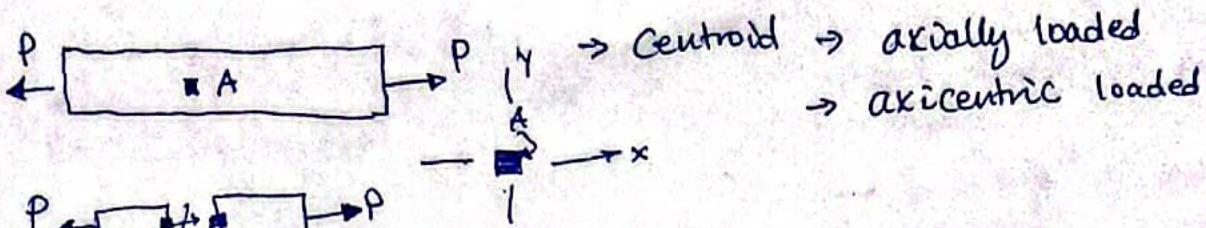
$$0.104 \times 10^{-6} \text{ m}^4$$

modulus of rigidity  $c = 0.02 \text{ m}^2$

$\gamma = \frac{\tau}{G}$  (gamma) (shear strain)  $\rightarrow$  angular deformation

$$P = 13.2 \text{ kN}$$

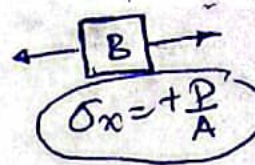
# Review of Strength of Materials



uniformly pressure.

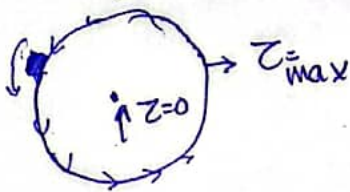
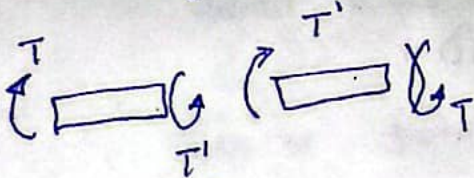
$$\text{load } (P) = \frac{\sigma}{A}$$

=

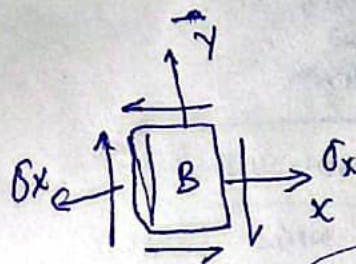


$$P = (\sigma)(A)$$

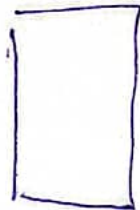
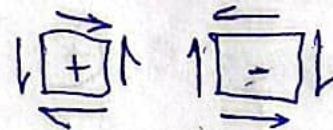
- temperature change
- determinate / indeterminate
- Hooke's law

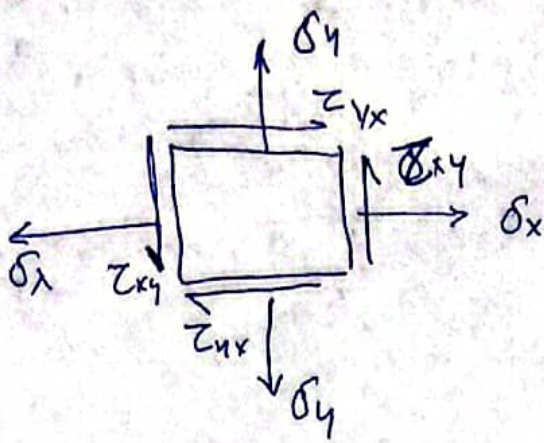


$$\tau = \frac{T \rho}{J}$$



$$\tau_{xy} = -\frac{T \rho}{J}$$



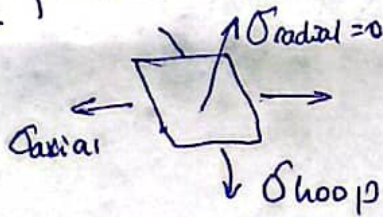


→ all

$$\sigma = \begin{bmatrix} \sigma_x & \tau_{xy} \\ \tau_{yx} & \sigma_y \end{bmatrix}$$

⇒ Plane stress conditions

- thin pressure vessel



- Teeth of spare gear if width is narrow

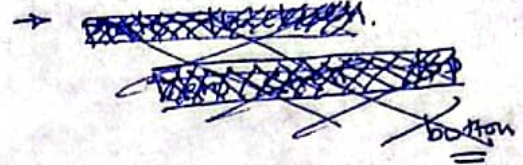
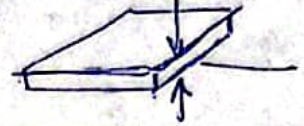
Stress transformation:

3D - 2D

8 plane stress strain

→ all stresses are on a same plane

→ important to analyse thin components



→ normal and shear stresses at free surface are zero

⊕ - since thin  
No variation in stress level

→ Only non zero stresses are in the x & y

∴ Plane stress condition is applied.

= Engineering Judgement

since  $\sigma_z \neq 0$

$$\theta = \tan^{-1} \frac{\tau}{\sigma}$$